



Lecture 6: Word Processing



Xu Ruifeng

Harbin Institute of Technology, Shenzhen



Last Time

- Language Modeling
- Statistical Language Model
- Neural Language Model



Today's class

- Chinese Word Segmentation
 - Problem Statement
 - Automatic Word Segmentation Algorithms
 - Benchmarks
 - Open Resources
- Unknown Word Identification and Normalization
 - Problem Statement
 - Unknown Word Identification
 - Unknown Word Normalization



Word Segmentation

- Definition

- A sentence S is a sequence of Chinese characters $(C_1, C_2, \dots, C_M)$. An n -character token, namely word, of S beginning at the i -th character is an ordered n -tuple of the form $(C_i, C_{i+1}, \dots, C_{i+n-1})$
- A partition W of a sentence partitions all its characters into non-overlapping tokens of various length $(C_1, C_2, \dots, C_{i_1-1})$, $(C_{i_1}, C_{i_1+1}, \dots, C_{i_2-1})$, $(C_{i_{N-1}}, C_{i_{N-1}+1}, \dots, C_M)$.
- Name the elements of W as $(W_1, W_2, \dots, W_N)$, respectively.

- Example:

- Input: character sequence
 - 他是研究物理的科学家。
- Output: word sequence
 - 他 是 研 究 物 理 的 科 学 家。



Why We Need Word Segmentation

- Accurate word segmentation is the foundation of Chinese language processing
 - Information Retrieval
 - 和服 | 务 | 于三日后裁制完毕，并呈送将军府中
 - 王府饭店的设施 | 租 | 服务 | 是一流的。
 - Text to Speech
 - 他们是来 | 查 | 金泰 | 撞人那件事的。“查” cha
 - 行侠仗义的 | 查金泰 | 远近闻名。“查” zha
 - Machine Translation
 - 我看见周星驰同张学友打招呼
 - Transtar: *I see week star Chi open together study friend greet.*



Segmentation Difficulties

- 这样|的|人才|能|经受|住|考验
- 这样|的|人|才能|经受|住|考验
- 白|天鹅|在|水|中|游来游去
- 白天|鹅|在|水|中|游来游去
- 我|将来|要|上|大学
- 我|将|来|上 海
- 两|个|人|一起|去
- 这|是|个人|问题



Segmentation Difficulties

- 今天|学生|会|面讨论|这个|问题
- 他|是|学生会|主席
- 这|篇|文章|太|平淡|了
- 难得一个|太平|盛世
- 莱温斯基|本来|就|很|不|爽
- 莱温斯|基本|来|就|很|不|爽
- 夏尔·莫里斯·塔列朗|的|祖先|从|10世纪|卡佩王朝|建立|时|起|就|已经|是|宫廷|贵人|了。他|的|父亲|塔列朗伯爵|查理-达尼埃尔|同|国王|路易十六|还|是|表兄弟



Two Main Challenges

- Two main Challenges
 - Ambiguity
 - Unknown word (Out of Vocabulary words, OOV)
- Types of ambiguity
 - Overlapping ambiguity (交叉型歧义)
“人才|能” and “人|才能”
 - Combinational ambiguity (组合型歧义)
“个人” and “个|人”
 - Mixed ambiguity (混合型歧义)
“太平|淡” and “太|平淡”



Out-of-vocabulary (OOV) problem

- Words unknown to dictionary “莱温斯基” “铁西”
- They are typically Named Entities 命名实体
 - Person name, e.g. “温家宝” “查理-达尼埃尔”
 - Location name , e.g. “马甸”
 - Organization name , e.g. “微软”
 - Informal word or new word in Internet, e.g. “欺实马” “亲们” “酱油帝”
- The unknown words caused the word segmentation accuracy loss at least five times more than word ambiguity [Huang and Zhao 2007]



National Standard

- 信息处理用现代汉语分词规范
 - GBT 13715-1992
- 刘源, 谭强, 沈旭昆 《信息处理用现代汉语分词规范及自动分词方法》
- 《资讯处理用中文分词规范》 台湾中研院
- 《人民日报》 语料库词语切分规范



Tasks of Typical Word Segmentation Module

- Dictionary lookup to get words
- Recognition of 重叠词、离合词 and 词缀 (SC*)
 - 重叠词: 高兴 → 高高兴兴, 高兴高兴
 - 离合词: 担心 → 担什么心, 洗澡 → 洗了个热水澡
 - 词缀: 标准 → 超标准, 科学 → 科学家
- Disambiguation in word segmentation
- OOV word detection
 - Named entity recognition (命名实体识别)



Word-based Approach: Dictionary-based

- Forward Maximum Matching method (FMM)
 - Greedy search routine
 - Find the longest string starting from the very point in the sentence that matches a word entry
- Backward Maximum Matching method (BMM)
 - Greedy search routine
 - Similar to FMM, but backward
 - More accurate



Word-based Approach: Dictionary-based

- Optimized Maximum Matching
 - Optimize the entries in the dictionary according to their frequency
 - Speed matching
 - No performance improvement
- Bi-directional Maximum Matching approach
 - applies FMM and then BMM
 - compares the two segmentation results to resolve any inconsistency and ambiguity
- Dictionary-based approach is difficult to process combinational ambiguities



Word-based Approach: Statistical-based

- Simple statistical approaches
 - [Sproat et al 2000] estimate the mutual information of two adjacent characters to determine whether they form a two-character word
 - [Sun et al 1998] further consider mutual information and the difference of t-score between characters
 - [Ge et al 1999]: a probabilistic model based on the Expectation Maximization (EM) algorithm
 - H1: There are a finite number of words of length 1 to k
 - H2: Each word has an unknown probability of occurrence
 - H3: Words are independent of each other
 - Method: Use EM algorithm to multi pass estimation
 - Word are the candidate multi-grams from training corpus
 - Word probabilities are randomly assigned initially
 - They are used to segment the text
 - The word probabilities are re-estimated based on segmented results
 - The text are re-segmented using re-estimated probabilities
 - Iterates until convergence

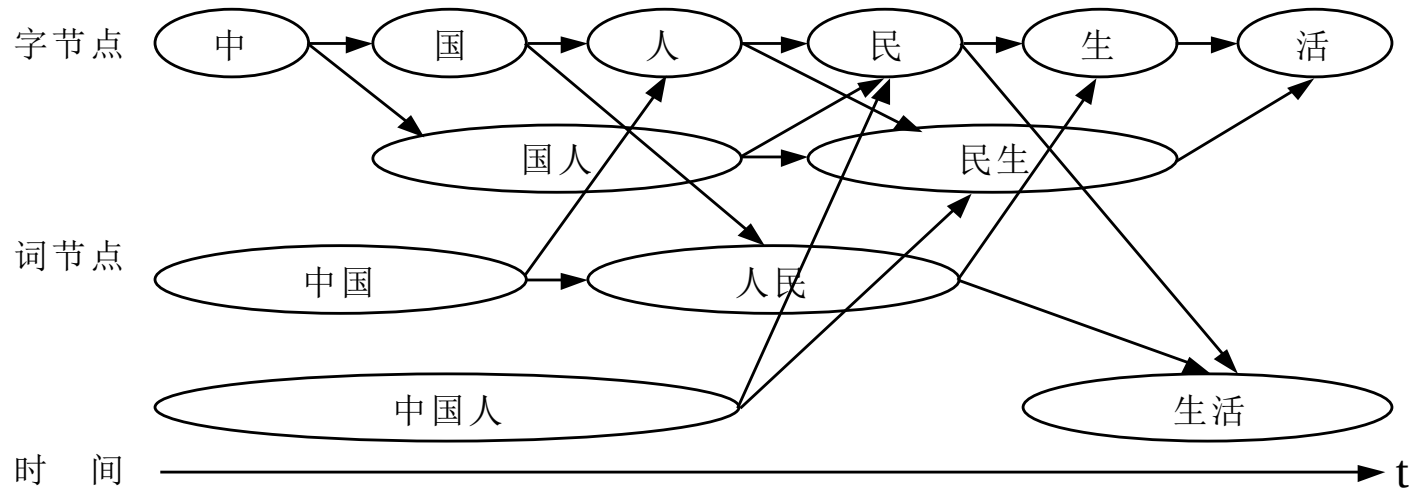


Machine Learning-based Approaches

- Transformation-based Algorithm
 - transformation-based learning (TBL) algorithm [Brill 1995]
 - requires a pre-segmented reference Corpus and an initial segmenter
 - the learning algorithm compares the initial segmentation with the reference Corpus
 - identifies a rule to correct the segmentation errors
 - initial segmentation is updated to outputs a set of ranked rules

The Word Lattice Based Approaches

- word lattice



- Apply different **machine learning based algorithms** to identify the **most likely path** in the word lattice
 - The shortest path
 - The maximum probability path
 - The maximum probability sum path
 - The maximum probability sum with shorter path
 - ...



Character-based Approach: Single Character-based

- Purely mechanical processes that extract certain number of characters (to form a string) from texts
- Divides Chinese texts into single characters
- Improved in Chinese Word Segmentation and Automated Indexing System (CWSAIS)
- Easy to implement
- Precision is low



Character-based Approach: Multi Character based

- Status of Chinese words

One Character	Two Character	Three Character	Four or More Character
5%	75%	14%	6%

- [Wu et al 1984]

- segment texts into strings containing two (bigram), three or more characters
- the bigram approach that segments a linear string of characters ABCDEF into AB, CD, EF and generates most of the correct Chinese word



Conditional Random Fields based

- Transfer word segmentation to Character labeling

上海 / 计划 / 到 / 本 / 世纪 / 末 / 实现 / 人均 / 国内 /
生产 / 总值 / 五千美元 / 。

上 / B海 / E计 / B划 / E到 / S本 / S世 / B纪 / E末 / S实
/ B现 / E人 / B均 / E国 / B内 / E生 / B产 / E总 / B值 / E
五 / B千 / M美 / M元 / E。 / S

B: the first character of a multi-character word

M: intermediate character in a multi-character word

E: the last character in a multi-character word

S: one character word

B / I / O

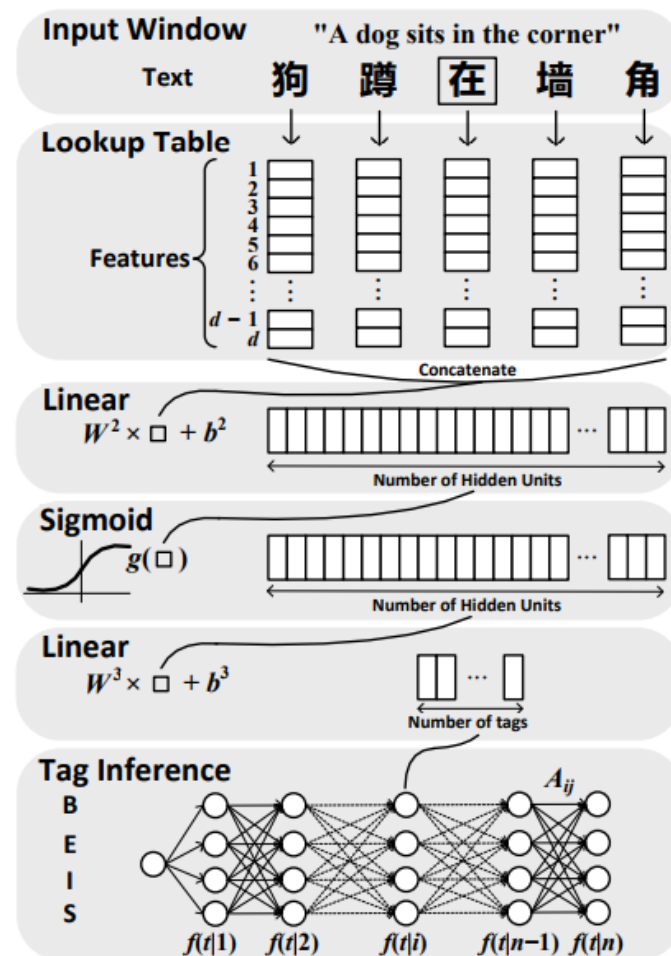


Conditional Random Fields based

- [Peng et al 2004, Tseng et al 2005, Zhou et al 2005]
Conditional Random Fields
 - CRF are undirected graphical models trained to maximize a conditional probability of the whole graph structure
 - CRF is a discriminative model which can capture many correlated features of the inputs
 - Suitable for sequence labeling
 - More accurate than the generative models
- Advantages of Character-based Approach
 - Simplicity and ease of application
 - Reduced costs and minimal overheads

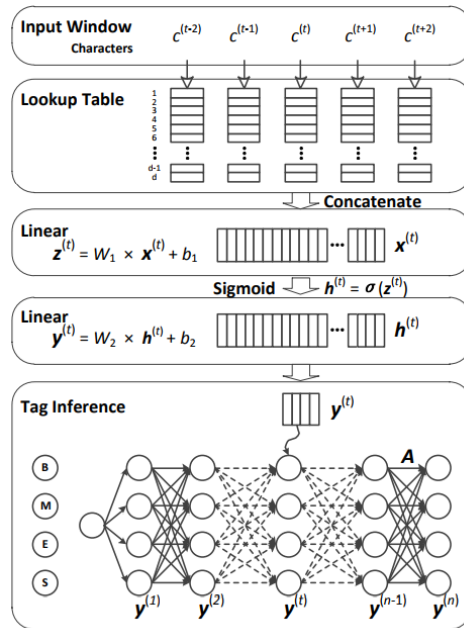
Neural-based Approach

- Sequence labeling schemes approaches
 - Zheng et al. (2013) first adapted the sliding-window based sequence labeling (Collobert et al., 2011) with character embeddings as input.

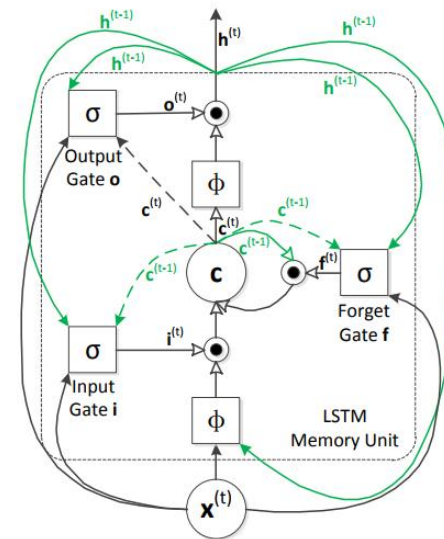


Neural-based Approach

- Sequence labeling schemes approaches
 - Chen et al. (2015b) used a Long shortterm memory network (LSTM) (Hochreiter and Schmidhuber, 1997) to capture long-distance context.



General architecture of neural model for Chinese word segmentation.

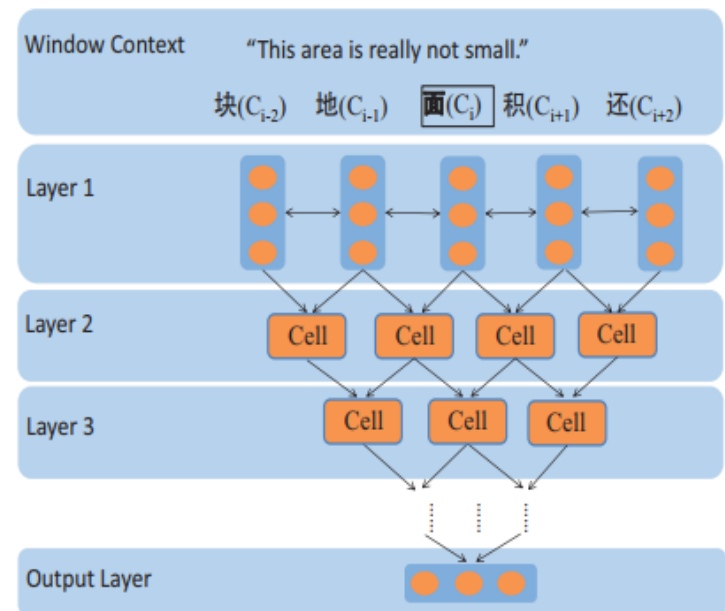


LSTM Memory Unit.



Neural-based Approach

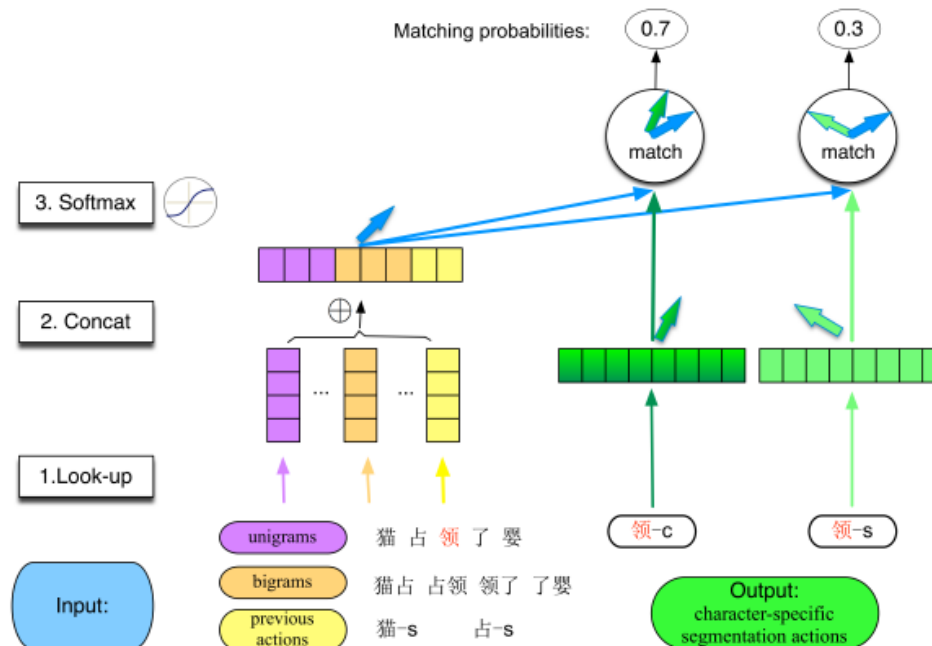
- Sequence labeling schemes approaches
 - Xu and Sun (2016) integrated both GRNN and LSTM for deeper feature extraction.
 - Called dependency-based gated recursive neural network(DGRNN).



Architecture of DGRNN for Chinese Word Segmentation. Cell is the basic unit of GRNN.

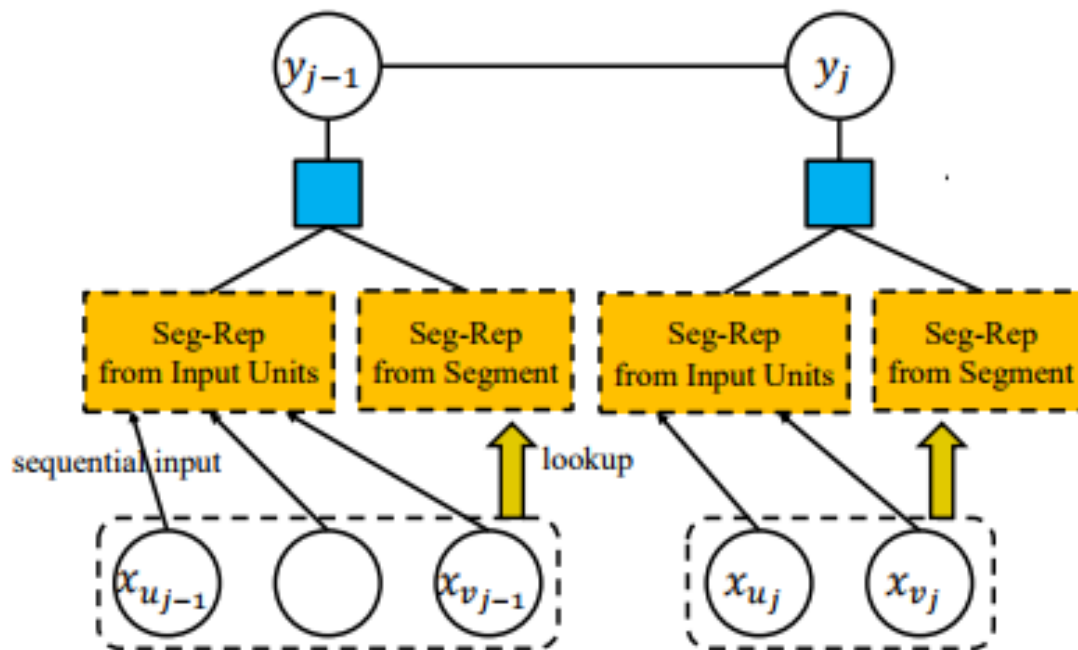
Neural-based Approach

- Other schemes approaches
 - Ma & Hinrichs (2015) proposed a character-based embedding matching approach.



Neural-based Approach

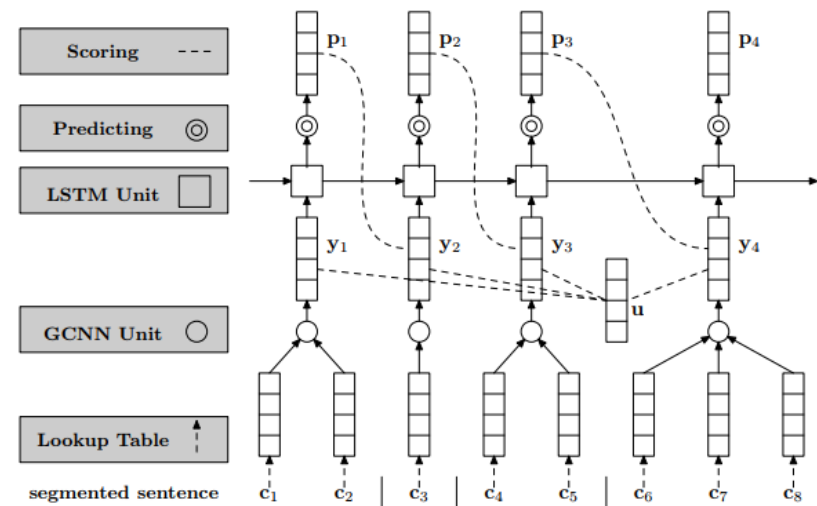
- Other labeling schemes approaches
 - Liu et al. (2016) used a zero-order semi-CRF based model.



neural semi-CRF model with segment representation from input composition and segment embeddings.

Neural-based Approach

- Other labeling schemes approaches
 - Cai and Zhao (2016) proposed to score candidate segmented outputs directly
 - Employing a gated combination neural network over characters for word representation generation and an LSTM scoring model for segmentation result evaluation.

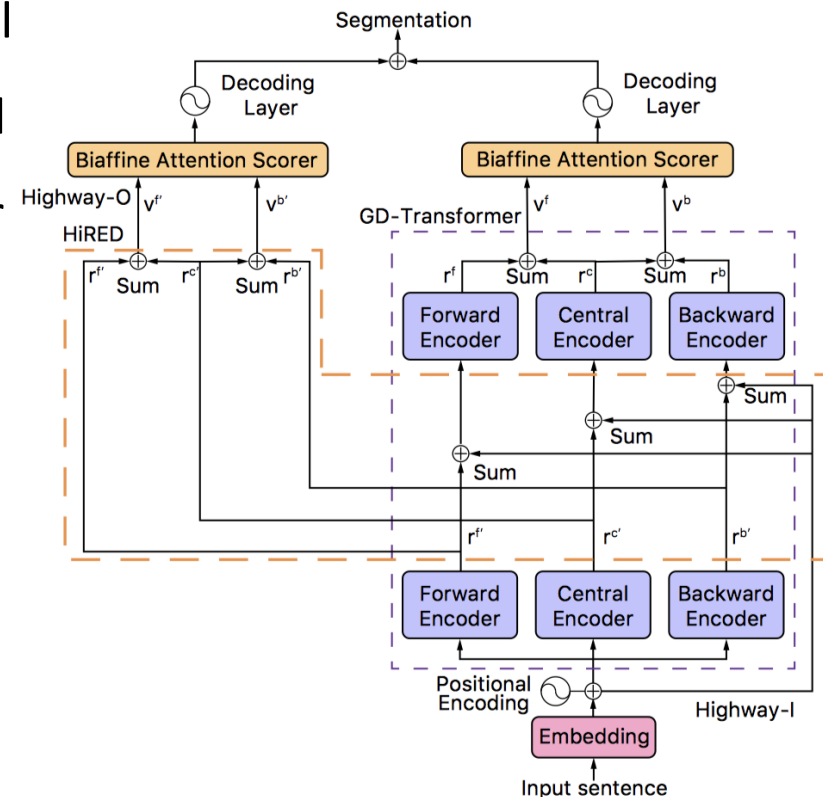


Architecture of proposed neural network scoring model



Neural-based Approach

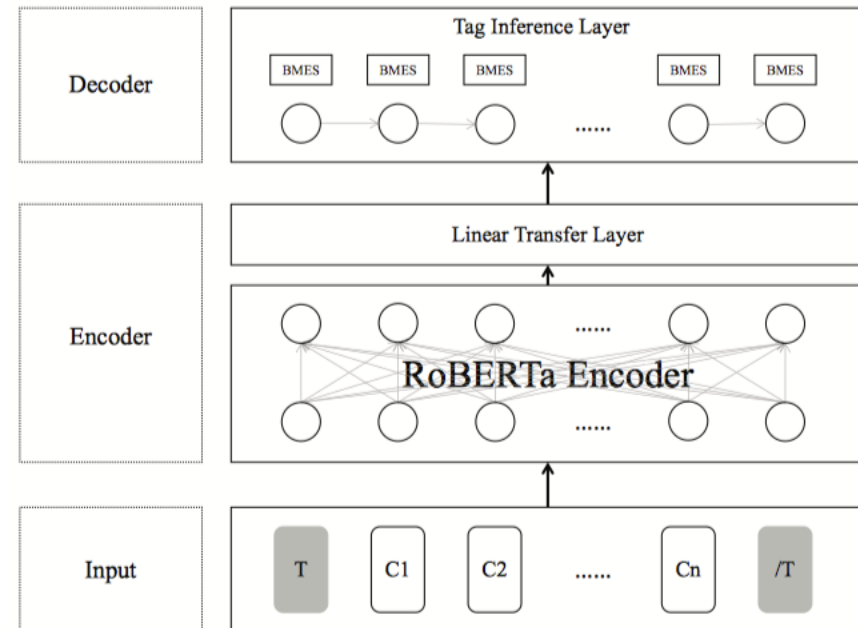
- Attention-based approach
 - Duan et al. (2020) proposed a model consists of an attention only stacked encoder and a light enough decoder for the greedy segmentation.
 - Encoder: represent the input
 - Decoder: perform actual segmentation





Neural-based Approach

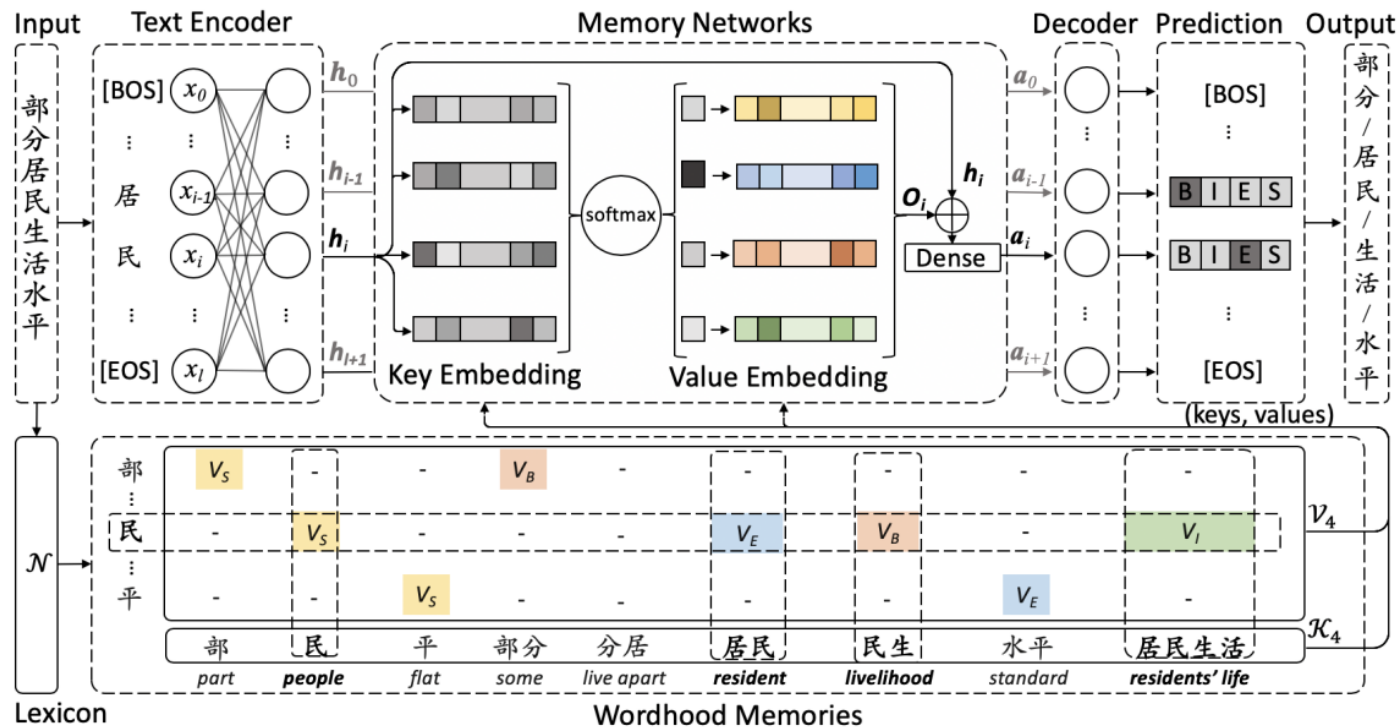
- Cross-domain related approach
 - Huang et al. (2020) propose a joint multiple criteria model that shares all parameters to integrate different segmentation criteria into one model.



Neural-based Approach

- General approach

- Ding et al. (2020) proposes a neural framework with a memory mechanism by leveraging wordhood information for general CWS.





Benchmark - Standards

- Peking University Standard
 - Contemporary Chinese Corpus
 - Characterized by a large set of specific rules
 - Each rule explains how a particular kind of character string should be segmented
 - Grammatical Knowledge-Base of Contemporary Chinese: 73,000 entries
 - Used as a reference lexicon



Benchmark - Standards

- Academia Sinica Standard
 - Sinica Corpus developed by Academia Sinica
 - Assumes the use of a reference lexicon for segmentation
 - No large set of specific rules
 - Two segmentation principles and four specific segmentation guidelines
 - A segmentation unit is defined as the smallest string of characters with an independent meaning and a fixed grammatical category



Benchmark - Standards

- University of Pennsylvania Standard
 - Similar to the Peking University Standard in the sense
 - Also consists of a large set of specific rules.
 - Each rule specifies how a particular kind of character string should be handled
 - Do not assume the use of any reference lexicon for segmentation.
 - Needs to determine the wordhood status for each character string without lexicon
 - The rules in the standard attempt to cover all possible scenarios, and the set become inevitably large



Benchmark - Bakeoffs

- To compare the accuracy of various method
- The “Bakeoff evaluation” first held at the 2nd SIGHAN Workshop at ACL in 2003
- Four corpora
 - Academia Sinica, Hong Kong CityU, Upenn, PKU
- Open Track
 - Allowed to use any other resources such as dictionaries or more training data
- Close Track
 - Only training data is allowed



Bakeoff Evaluation

- SigHan Bakeoff are evaluated in five measurements
 - recall, precision and F-measure for overall segmentation
 - recall for unknown words and known words

$$\text{Recall} = \frac{\text{number of correctly segmented words}}{\text{total number of words in gold data}}$$

$$\text{Precision} = \frac{\text{number of correctly segmented words}}{\text{total number of words segmented}}$$

$$F\text{-measure} = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$$

$$\text{Recall(OOV)} = \frac{\text{number of correctly segmented unknown words}}{\text{total number of unknown words in gold data}}$$

$$\text{Recall(IV)} = \frac{\text{number of correctly segmented known words}}{\text{total number of known words in gold data}}$$



4th Bakeoff Evaluation

- New Tasks
 - Chinese Word Segmentation
 - Chinese Named Entity Recognition
 - Chinese POS-tagging
- Seven Corpora
 - Academia Sinica, Hong Kong CityU, Microsoft Research Asia, PKU, Shanxi University, National Chinese Corpus, Chinese Tree Bank



Open Resources

- NLPIR ICTCLAS
 - Beijing Institute of Technology
 - <http://ictclas.nlpir.org/>
- JIEBA Word Segmentation System
 - <https://github.com/fxsjy/jieba>
- HIT LTP Platform
 - www.ltp-cloud.com/

Unknown Word Identification and Normalization



- Problem Statement
- Typical Features and Techniques
- Unknown Word Identification
 - Chinese Person Name Identification
 - Chinese Organization Name Identification
 - Chinese Place Name Identification
- Unknown Word Normalization
 - Abbreviation



Unknown Word

- Out Of Vocabulary (OOV) problem
 - Rapid growing new words 非典 给力 国考
 - OOV caused errors are about 5 times to ambiguity caused errors
- Five major types :
 - Abbreviation (acronym)
 - 国考=国家公务员考试
 - Proper name/Named Entity
 - Names of people PG One
 - Names of places 延坪岛
 - Names of organizations 上海合作组织
 - Certain key words are indicators for each different sub-category
 - Derived words:
 - Have affix morphemes which are strong indicators
 - 审计 -> 审计人 审计员 审计局 审计处
 - Compounds:
 - A word made up of other words
 - 光+敏感 -- 光敏感 流体+力学 – 流体力学
 - Numeric type compounds:
 - 五月三日 八点十分 第一

Unknown Word Detection and Recognition



- Decide whether an OOV string can form a word or not
 - Rule-based approach
 - Compositional Rule
 - Reject Rule
 - Statistics-based approach
 - In-word probability
 - Boundary probability
 - Learning-based approach



Typical Features and Techniques

- Frequency/Relative Frequency
 - Starting point to pick up new word candidates
 - low frequency new words are hardly identifiable
- In-word probability
 - Reflect the morphological property of Chinese
 - [Chen et al 2005; Wu et al 2000; Fu 2001] used in-word probability to combine adjacent single characters after initial segmentation
- Word Formation Pattern [Wu et al 2000]
 - Describe how likely a character appears in a certain position within a word
- Mutual information [Church et al 1990]
 - Estimates the internal association strength among constituents of character n-grams

Chen A, Zhou Y, Zhang A, et al. Unigram language model for Chinese word segmentation.

Wu A, Jiang Z. Statistically-enhanced New Word identification in a rule-based Chinese system.

Fu G. Research on statistical methods of Chinese syntactic disambiguation.

Church kW, Hanks P. Word association norms, mutual information, and lexicography.



Typical Features and Techniques

- Left/right entropy [Sornlertlamvanich et al 2000] and context dependency [Chien 1999]
 - Describe the dependency strength of current item (character sequence) on its context
 - The dependency strength of a current item on its context decreases as the probability of this item to be a Chinese word increases
- Independent word probability [Nie et al 1995]
 - A property of a single character or a string of characters
 - Likelihood for this character to appear as an independent word in texts $IWP(c) = \frac{N(Word(c))}{N(c)}$
- Anti-word list [Li et al 2004; Nie et al 95]
 - A list of functional characters to exclude bad candidates
 - Single character prepositional, adverbial and conjunctive words etc.

Chien LF. PAT-tree based adaptive keyphrase extraction for intelligent Chinese information retrieval.

Sornlertlamvanich V, Potipiti T, Charoenporn T. Automatic corpus-based Thai word extraction with the C4.5 learning algorithm.

Nie JY, Hannan ML, Jin WY. Unknown word detection and segmentation of Chinese using statistical and heuristic knowledge.

Li H, Huang CN, Gao J, Fan X. The use of SVM for Chinese new word identification.



Typical Features and Techniques

- Similarity between new words and dictionary words [Li et al 2004]
 - If two characters appear more times in the same word patterns, the analogy between them is more reliable
 - 下 (below) has the most identical word patterns with 上 (above), and there is a strong preference

$$ANA(a, x) = \frac{\sum_c (W(ac)W(xc) + W(ca)W(cx))}{\sum_c (W(ac) + W(ca) + W(xc) + W(cx))}$$

- “overlap variety”
 - In [Ye et al 2013] the likelihood of an character n-gram being a meaningful Chinese word is measured by “overlap variety”
 - a criteria which considers the goodness of the candidate together with the goodness of the strings overlapping the candidate.
 - The authors have also adopted a word refinement module to filter out known words during the post-processing.



Typical Features and Techniques

- [Mo et al 2016] maximized substrings
 - a simple yet effective approach for extracting a specific type of frequent substrings
 - provide good estimations of unknown word boundaries.
- [He et al 2017] propose a unified model which can learn from out-of-domain corpora and in-domain unannotated texts.
 - The unified model contains two major functions.
 - Cross-domain learning function can learn out-of-domain information based on domain similarity.
 - Semi-Supervised learning function can learn in-domain unannotated information by self-training.

Typical Features and Techniques

- [Rui et al 2016] proposes a method to detect unknown words during natural reading of non-native language text
 - using eye-tracking features
 - They propose several eye gaze features and classifiers to detect unknown words.

Table 2: Feature selection ranking by correlation coefficient r .

Feature	r	p -value
word rarity : $rare(w)$	0.462	e-233
word length : $len(w)$	0.339	e-171
max gaze duration : $g_{max}(w)$	0.318	e-53
first gaze duration : $g_{first}(w)$	0.265	e-15
total gaze duration : $g_{total}(w)$	0.252	e-45
number of fixation : $n(w)$	0.208	e-40
mean fixation : $f_{mean}(w)$	0.153	e-31
pupil variation: $p_{variation}(w)$	0.136	e-19
regression : $reg(w)$	0.109	e-16

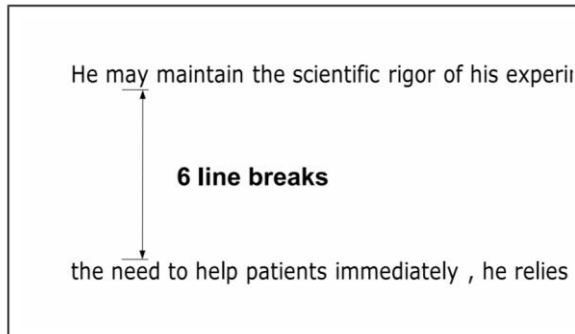


Figure 2: An example of the stimulus document.

Table 3: Classification results.

Features	$rare$ g_{total}	$rare$ g_{total}	$rare$ len	$rare$ g_{max}	$rare$ g_{max} len
classifier	Baseline	SVM	SVM	SVM	SVM
Precision	0.482	0.457	0.501	0.472	0.507
Recall	0.375	0.630	0.581	0.639	0.616
False Positive	0.027	0.054	0.042	0.052	0.043
F-measure	0.422	0.530	0.538	0.543	0.556

Neural-based Approach for OOV Word Embedding

- Few-Shot Learning OOV Words
 - Hu et al. (2019) formulate the learning of OOV embeddings as a few-shot regression problem, and address it by training a representation function to predict the oracle embedding vector (defined as embedding trained with abundant observations) based on limited observations.
 - propose a novel hierarchical attention-based architecture to serve as the neural regression function.

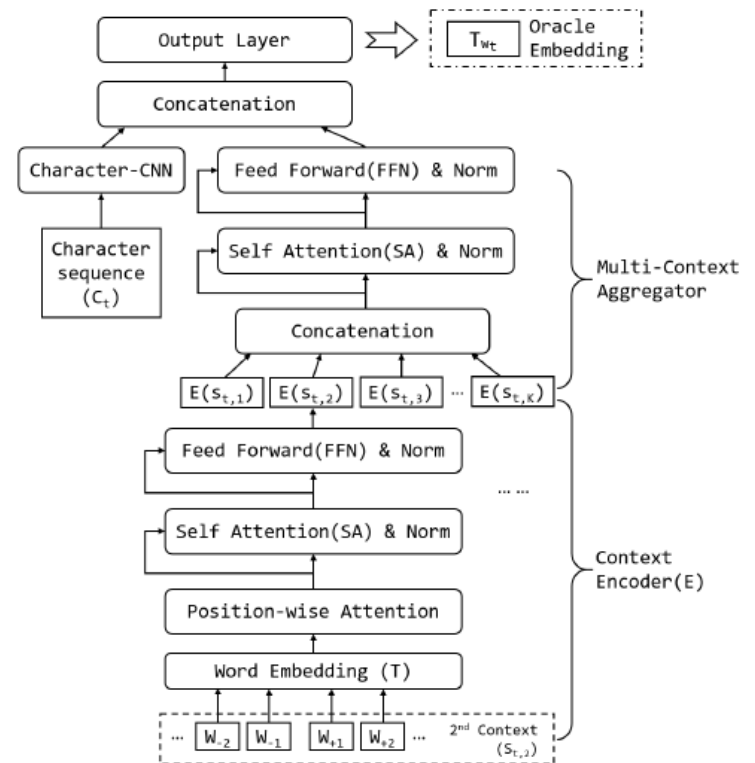
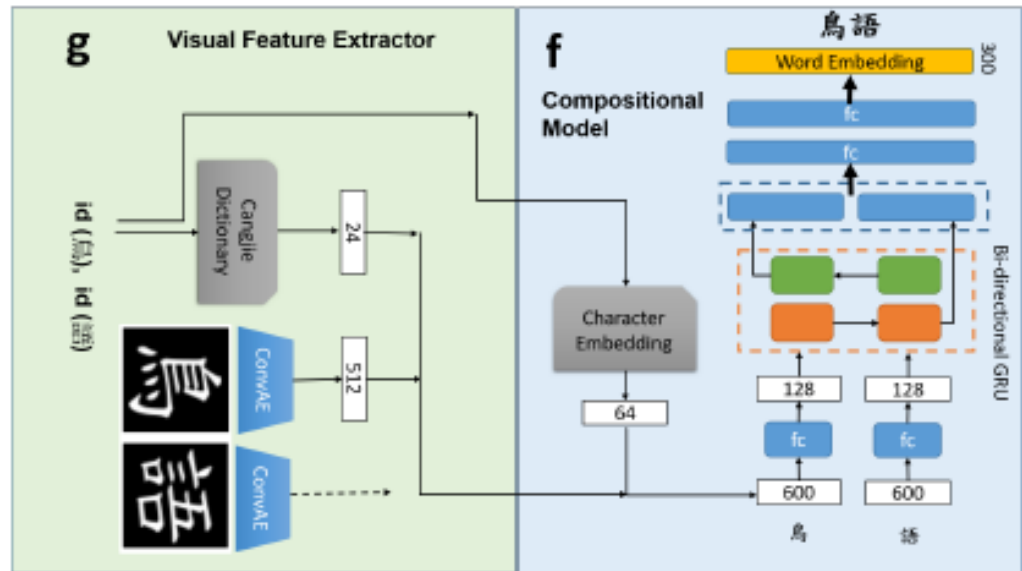
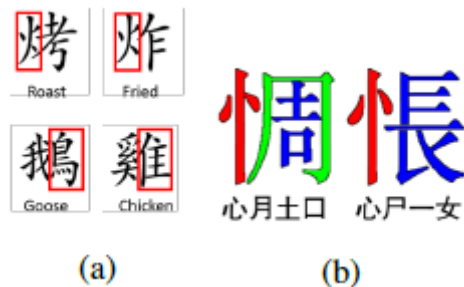


Figure 1: The proposed hierarchical context encoding architecture (HiCE) for learning embedding representation for OOV words.

Neural-based Approach for OOV Word Embedding

- Learning Chinese OOV word embedding from Glyphs
 - Chen et al. (2020) present a multi-modal model, Glyph2Vec, to tackle Chinese out-of-vocabulary word embedding problem.
 - Glyph2Vec extracts visual features from word glyphs to expand current word embedding space for out-of-vocabulary word embedding.





Chinese Person Name Identification

- A Chinese person name in Chinese is a family name + a given name
 - 李刚 朱五六 欧阳奋强 公孙杵臼
 - 25%+ out-of-vocabulary words
- Challenges:
 - Name formation is arbitrary
 - A name does not have “boundary tokens”
 - Different news agencies and people in different areas may translate the same foreign name to different Chinese names



Chinese Person Name Identification

- Driven by family name.
 - Less than 300 family names are commonly used
 - 千家姓 百家姓总汇 姓氏词典
- Detection of the name right boundaries

Types of Ordinary Words in Names	Examples
Two Character Name	高峰, 文静, 支流, 幸运, 关键
Three Character Name	黄灿灿
The First Two Characters	刘海亮, 黄金荣
The Last Two Characters	朱俊杰, 叶海燕



Chinese Person Name Identification

- Name identification using contextual information
 - Many studies constructed context word list
 - Title words: 总理/厂长
 - Verbs that are normally followed by person names, 授予 接见
 - Patterns that contain both left and right context words flanking the names: 任命 ... 为” and “记者 ... 报道”
- Radical is the primary component of a Chinese character
 - Semantic radicals
 - Conveys the range or some aspect of the meaning of the whole character 玉-璧 莹 瑜 琼 琪 玲 琅
 - Gender 薔 薇 / 杏 桃
 - Contemporary characteristics 李援朝 王建国
 - Political Taboo 李世勣 → 李勣 党怀英
 - Phonetic radicals
 - Often related to the pronunciation of the whole character



Transliteration of Foreign Names

- Foreign names are usually transliterated from the pronunciations in the original languages : Google, Facebook, Twitter
- Can be any lengths
- Features:
 - Chinese characters that appear particularly often in person name transliterations 尔 姆 斯
 - Boundary characters/string 夫、斯基
 - Structural feature 阿历克谢.马克西莫维奇.彼什科夫
 - Contextual information



Case Study:

HMM Based Person Name Identification

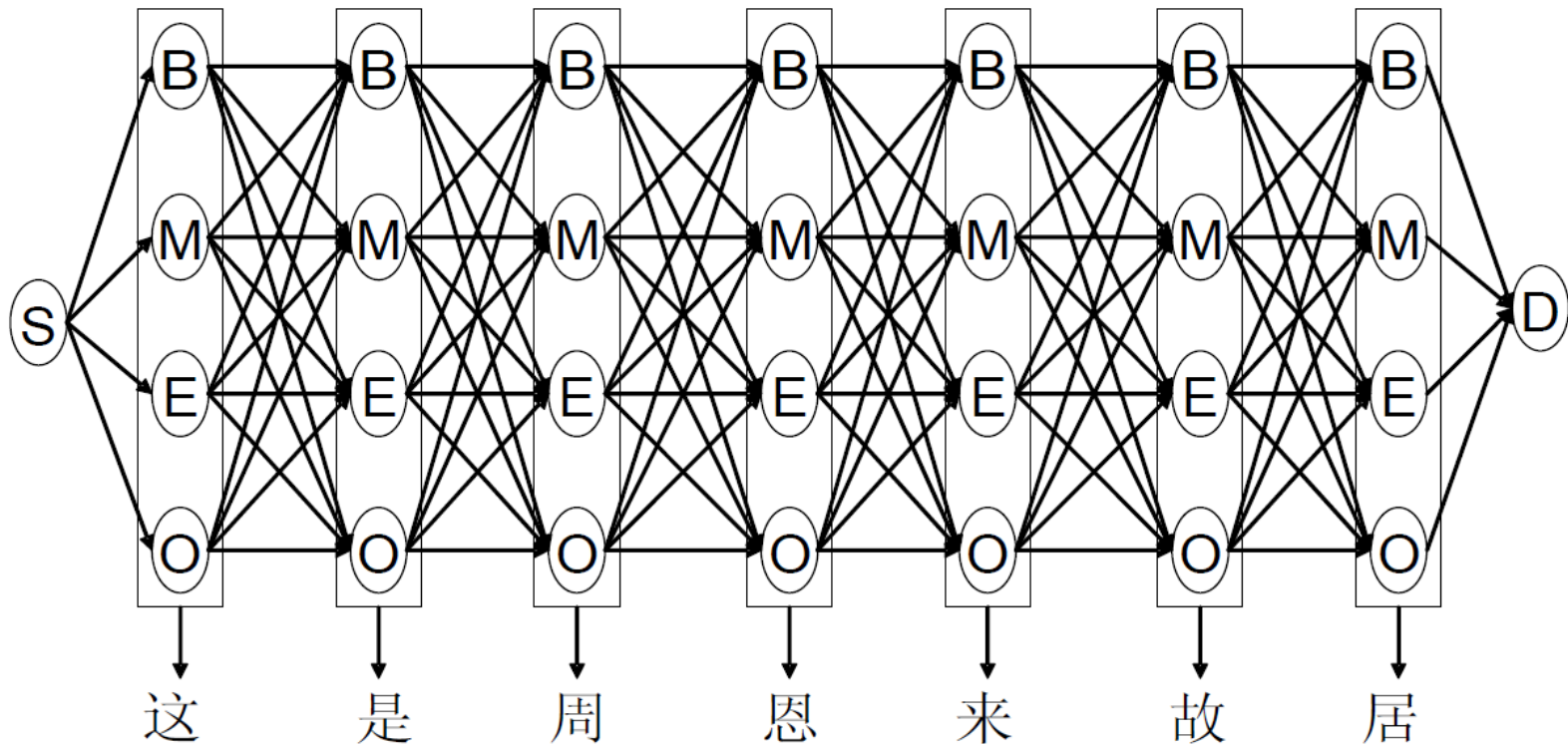
- Transfer name identification problem into tagging problem:
 - Input text:

这是周恩来、邓颖超生前居住的地方
 - Tagged result:

这是周恩来、邓颖超生前居住的地方
O O B M E O B M E O O O O O O
 - O: not OOV, B: The first character of OOV
 - M: The middle character of OOV E: The last character of OOV
- "周恩来" and "邓颖超" which were tagged to "BME" are identified as OOV
- Use OOV tagged corpus as training corpus

HMM Based Person Name Identification

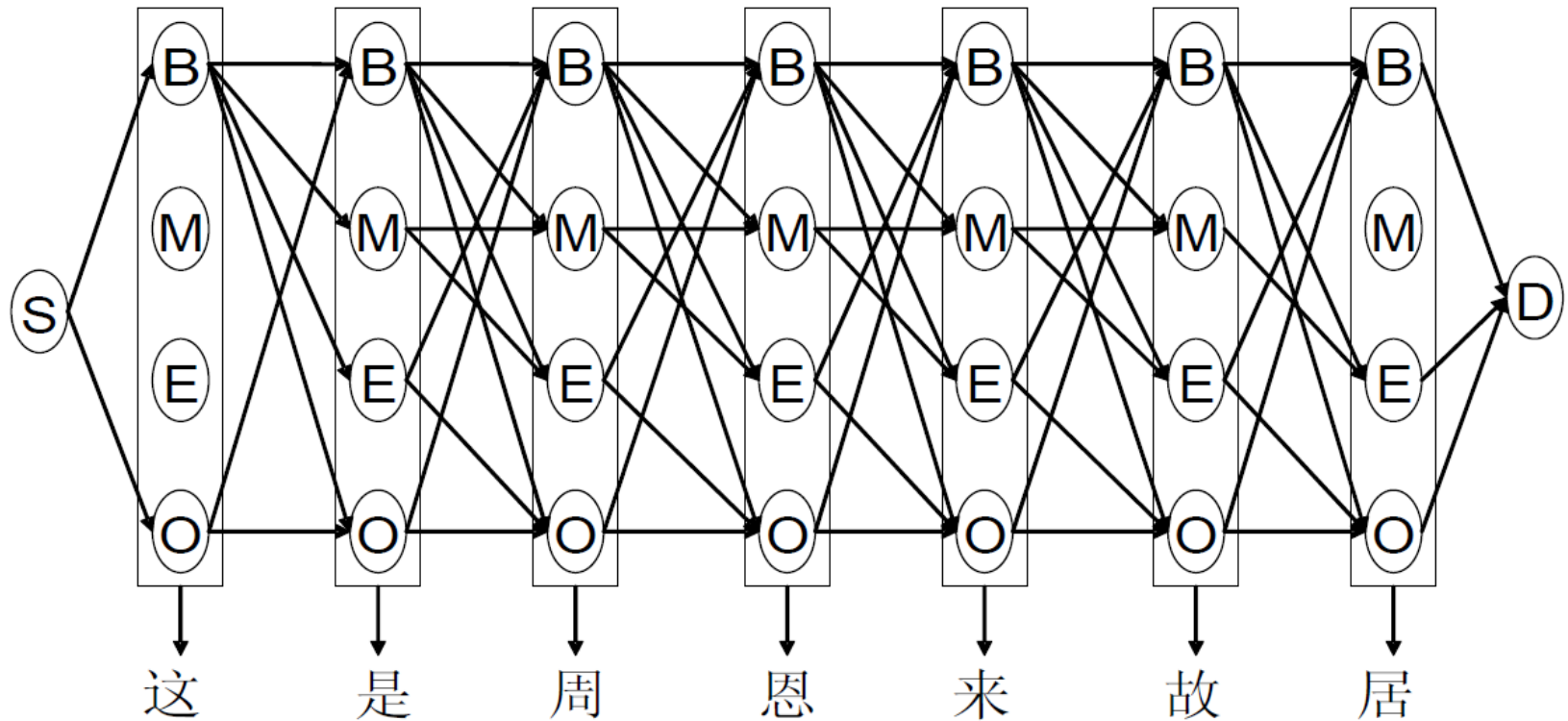
- Generate all of the edges



-

HMM Based Person Name Identification

- Remove unreachable edge:





HMM Based Person Name Identification

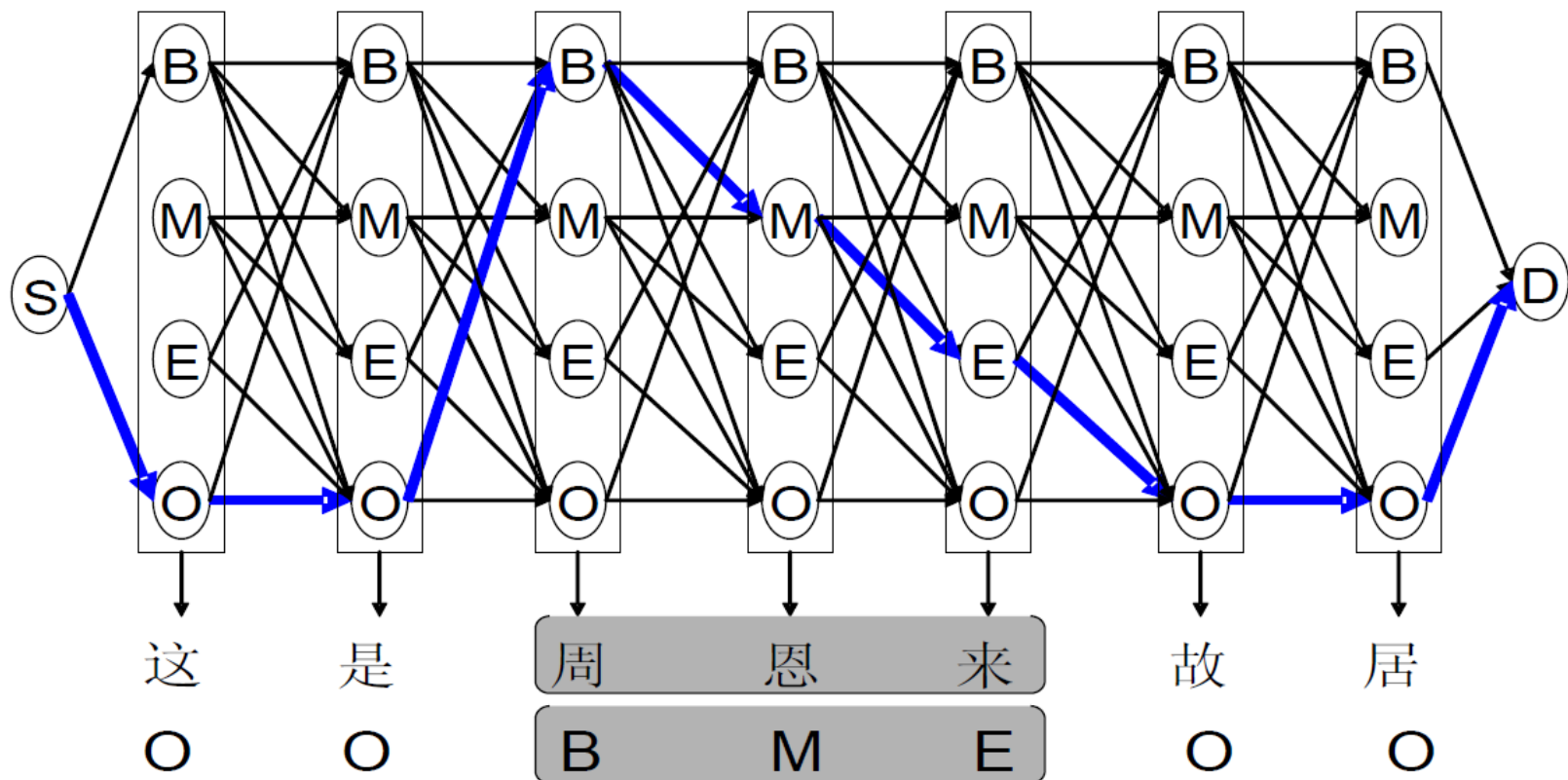
- Match the following tagging segments in the optimal tagging path:

- B
- BE
- BME
- BMME
-

} 人名标记序列

HMM Based Person Name Identification

- Find person name segment in the tagging sequence:





Chinese Organization Name Identification

- Organization name composition are complex
 - The only clear clues are the organization types
- Have no predictable length
- Embody many other types of names
 - 福田区委 李嘉诚慈善基金会
- Words included in organization names are diverse
- Many organization names have abbreviations
 - 微软亚洲研究院



Internal Structure of Chinese Organization Name

- A Chinese organization name is normally composed of two major components
 - The first part is the proper name: Prefix word
 - Unlimited: 光明区委 三一重工集团
 - The second part is the organization type: organization name characteristic words
 - 公司 基金会 大学
 - The second part contains the major clue words guiding the identification of the right boundary
 - Determining the left boundaries becomes most crucial to organization name identification.



Features

- Features for organization name composition
 - Probability of word/character in/out organization name
→ organization word/character list
 - Famous organization names and their abbreviations
- Features for potential left boundary
 - Location Name 杭州商会
 - Organization Name 哈尔滨工业大学深圳研究生院
 - Special Word: NBA中国球迷会



Constraints For Organization Name Composition

- Some “stop words” such as 失败 and 其它 will never be selected to be part of the name
- Rule-based approaches[Chen et al 2000; Zhang et al 2008]
- Probability-based approaches [Yu et al 2003; Sun et al 2003; Wu et al 2003]
- Role-based tagging approach [Yu et al 2003]
- Web-based approach

Chen KJ, Chen CJ. Knowledge extraction for identification of Chinese organization names.

Yu H, Zhang H, Liu Q. Recognition of Chinese organization name based on role tagging.

Sun J, Zhou M., Gao J. A class-based language model approach to Chinese named entity identification.

Wu Y, Zhao J, Xu B. Chinese named entity recognition combining a statistical model with human knowledge.

Zhang Q Hu G, Yue L. Chinese organization entity recognition and association on web pages.



Chinese Place Name Identification

- Place name dictionary and dictionary look-up
 - The Chinese Place Name Set
 - published by Place Names Committee of China
 - 100,000 place names
 - 中华人民共和国地名词典
 - 中国古今地名大词典
 - 中文地名索引
- About 30% of the places names in a real text cannot be found [Zhang et al 2001]



Chinese Place Name Identification

- Combine corpus statistics and context rules
- The place suffix is a kind of useful features
- Corpus-based approaches:
 - Estimate the likelihood of a character being a part of a Chinese place name
 - Used as the beginning, middle and end character
 - Capture the capability of a character forming for Chinese place names
 - Provide a good means to locate the place name candidates



Chinese Place Name Identification

- Rule-based Candidates Confirmation

- Example Rule: If two place name candidates are coordinated and one of them has been confirmed as a true place name, then the other should be confirmed as a true place name also
- E.g. [罗湖区]、[福田区]和光明新区

- Rule-based Candidates Elimination

- Example Rule: A place name candidate should be eliminated if its preceding word is a title of a person
- E.g. 王四川 小沈阳



Abbreviation

- The first character of the each word in full name,
哈尔滨工业大学 → 哈工大
- A proper name in an organization full name, select the proper name
耐克公司 → 耐克
- Location name plus the first character of the other words
上海交通大学 → 上海交大
- First character of all the words excluding the organization name characteristic word.
中国交通银行总部 → 交行总部



Abbreviation in Chinese

Given a full-form $F = f_1 f_2 \cdots f_m$ and its corresponding abbreviation $S = s_1 s_2 \cdots s_n$, let $f_i (1 \leq i \leq m)$ denote a constituent word of the full-form and $s_j (1 \leq j \leq n)$ denote one component of the relevant short-form, then the above three types of Chinese abbreviations can be formally redefined as follows:

- Reduced Abbreviation: 空军政治部 – 空政

If $n = m$ and s_i is the corresponding short-form of the constituent word f_i

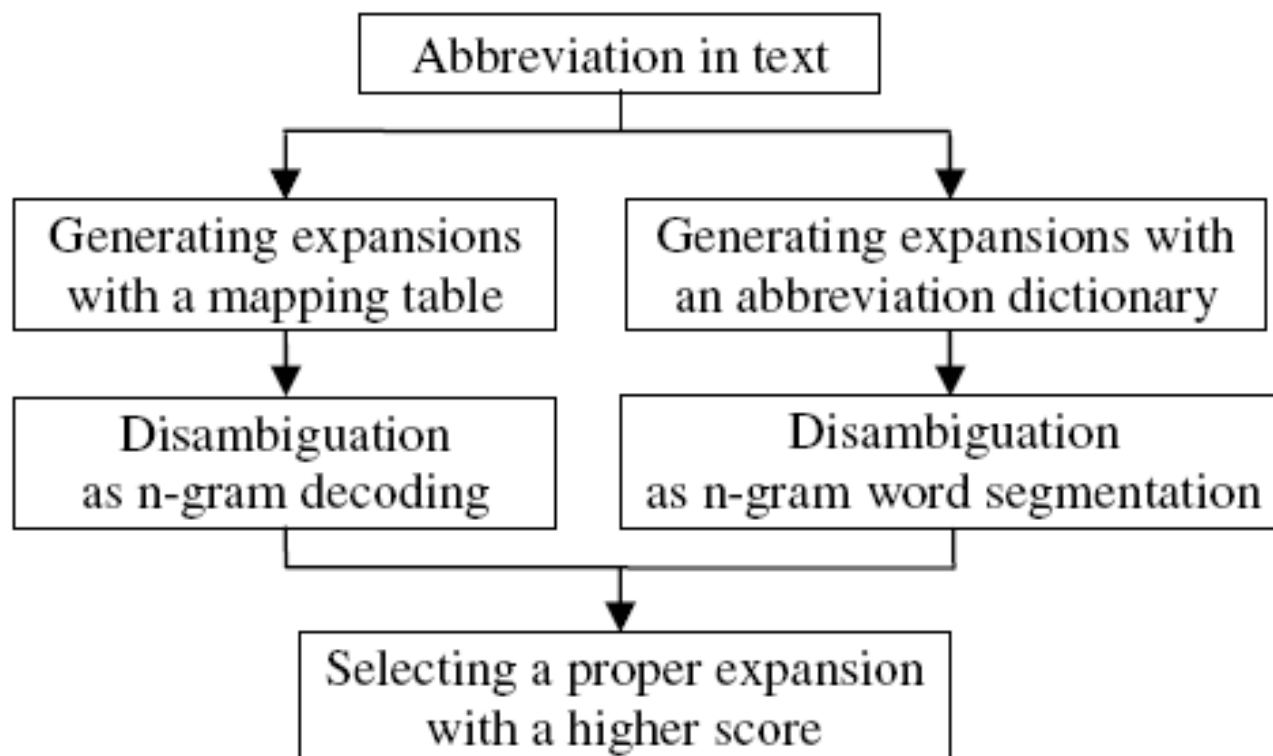
- Eliminated abbreviation: 清华大学- 清华

If $n < m$ and $\forall s_j \in S (1 \leq j \leq n)$

- Generalized abbreviation: 三从四德 八荣八耻

If $n < m$ and $\exists s_j \notin F (1 \leq j \leq n)$

Chinese Abbreviation Normalization: Framework





Expanding Reduced Abbreviations

- Generating Expansions with a Mapping Table
 - Assumption: Each character or word in a reduced abbreviation must match a word in its full-form
 - Characters or short-words vs. Full words

Short Words	Full Words	Real Cases
大	大学 大会 大型	北大 十五大 十大工程
联	联邦 联合会	足联 国联



Expanding Reduced Abbreviations

- Disambiguation as N-Gram Decoding
 - Expansion decoding aims to search a combination of long-words as the best path from the lattice that maximizes the n-gram probability

$$\begin{aligned}\hat{F} &= \arg\max_F P(LFR) = \arg\max_F P(Lf_1f_2 \cdots f_nR) \\ &\approx \arg\max_F P(f_1 | L) \times \prod_{i=1, n} P(f_i | f_{i-N+1, i-1}) \times P(R | f_{n-N+2, n})\end{aligned}$$

$$\hat{F} = \arg\max_F P(Lf_1f_2 \cdots f_nR) \approx \arg\max_F \prod_{i=1}^{n+1} P(f_i | f_{i-1})$$



Expanding Non-Reduced Abbreviations

- Generating Expansions with a Dictionary of Abbreviations
- Maps each non-reduced abbreviation to a set of full-forms from a abbreviation-expanded corpus

Short Words	Full Words
清华	清华大学
三通	通邮、通商、通航



Expanding Non-Reduced Abbreviations

- Disambiguation as N-Gram Word Segmentation
 - Select a proper expansion from a set of candidates if any
 - Using bigram word segmentation
 - The main idea is: each expansion candidate of a given abbreviation is segmented into a sequence of words using bigram LMs. The one whose segmentation has the maximum score will be identified as expansion output



Abbreviations Expansion Disambiguation

- Final disambiguation for expansion candidates
- Comparing their respective n-gram scores
- The one with a higher score is chosen as the resulting expansion

- Performance 85.5-88.0%
- Problem Left
 - 志愿: 北大 科大 清华
 - 上吊 自砂



Automatic Abbreviations for Titles

- Abbreviations are simply the combination of the first characters of all words in the title(example 1)
- Choosing to combine prefixes of certain words in the title (example 2-3)
- Using some external words (example 4)

Id	Abbreviation	Title Words
1	FAWN	A Fast Array of Wimpy Nodes
2	IPR	An Integrated Placement and Routing Algorithm
3	SeRQL	A Second Generation RDF Query Language
4	SIMPLicity	Semantics-Sensitive Integrated Matching for Picture Libraries

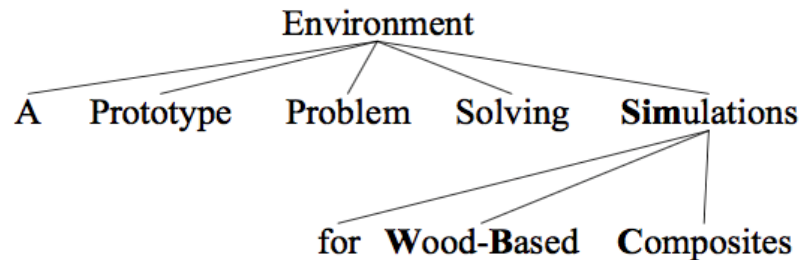


Description Analysis

- Lexical Analysis (Lex)

$$f_{\text{Lex-X}}(w_k) = \begin{cases} 1 & \text{the POS tag of } w_k \text{ is X,} \\ 0 & \text{otherwise.} \end{cases}$$

- Syntactic Analysis (Syn)



- Semantic Analysis (Sem)

$$f_{\text{Syn-i}}(w_k) = \begin{cases} 1 & w_k \text{ is at the } i^{\text{th}} \text{ layer of the tree,} \\ 0 & \text{otherwise.} \end{cases}$$



Candidate Generation

- Indicator features

$$f_{\text{Char-}i}(p_k, w_k) = \begin{cases} 1 & p_k \text{ uses the } i^{\text{th}} \text{ character of } w_k, \\ 0 & \text{otherwise.} \end{cases}$$

- Score of the prefix

$$s_{\text{prefix}}(p_k, w_k) = s_{\text{word}}(w_k) \times \vec{w}_{\text{Char}} \cdot \vec{f}_{\text{Char}}(p_k, w_k)$$

- Score of the Candidates

$$s_{\text{ABBR}}(a) = \frac{1}{l} \sum_{k=1}^n s_{\text{prefix}}(p_k, w_k)$$



Abbreviation Ranking

- Abbreviations that are, or similar to, existing words are usually good candidates,
- Denote the language model score of abbreviation:

$$s(a) = s_{\text{ABBR}}(a) + w_{\text{LM}} \times f_{\text{LM}}(a) + w_{\text{len}} \times f_{\text{len}}(a)$$

- Performance 50.0-55.0%
- Problem Left
 - Do not consider sophisticated language phenomena



你知道“小火车”是中国哪一型军舰的昵称吗？



“小火车”是中国海军的“江湖级”导弹护卫舰的昵称。这个昵称源自该型舰船的外形，其船体轮廓在一些人眼中类似于小型火车的车厢。江湖级导弹护卫舰是中国自主研发的一种护卫舰，以其高性价比和灵活多变的作战能力而闻名。

为什么叫这型军舰为小火车？



“小火车”这一昵称是因为江湖级导弹护卫舰的船体轮廓在某种程度上与小型火车的车厢相似。这种军舰的船体呈长方形，相对于一些其他军舰，它的船体相对较矮、较长，有时被认为与火车车厢的形状相近，因此得到了这个昵称。这种称呼可能是中国海军内部或舰队之间的俚语或幽默用语，用于形象地描述这种型号的舰船外观。



055型驱逐舰是中国最新的大型导弹驱逐舰，请给这型军舰起个昵称



“华山级”



你知道577是哪个欧洲国家的领导人吗？



抱歉，我无法提供当前时间点的特定领导人信息。577并不是一个我熟悉的数字或者国家代码。可能是一个笔误或者误解。如果你能提供更多信息，我可以尽力帮助你找到答案。

🔊 📄 ↺ 🍏 📌 ✨

你知道577是哪个欧洲国家的领导人的别名或者昵称吗？



抱歉，我不知道任何欧洲国家的领导人使用"577"作为别名或昵称。如果这是一个特定人物的别名或昵称，可能是一种特定的个人称呼，或者是某个特定社区或团体内的称呼，不过这并不是一个我所熟知的信息。



The next lecture

- Lecture 7
Word Meaning and Word Sense
Disambiguation